ITNET03 Network Design Project
Phase 2
Alonzo IT Training Center Network Documentation

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1. IP Addressing Scheme

1.1 Subnet Table

Subnetwork	VLAN	Network Address	Subnet Mask	Host Address Range	Notes
STUDENTS	*10*	*172.16.0.0*	*255.255.255.0*	*172.16.0.1 - 172.16.0.254*	*Supports 254 hosts Est 120 hosts*
FACULTY	*20*	*172.16.1.64*	*255.255.255.192*	*172.16.1.65 - 172.16.1.126*	*Supports 62 hosts Est 26 hosts*
OPERATIONS	*30*	*172.16.1.128*	*255.255.255.192*	*172.16.1.129 - 172.16.1.190*	*Supports 62 hosts Est 28 hosts*
IT_OFFICE	*40*	*172.16.1.224*	*255.255.255.224*	*172.16.1.225 - 172.16.1.254*	*Supports 30 hosts Est 13 hosts*
SERVICES	*50*	*172.16.2.0*	*255.255.255.240*	*172.16.2.1 - 172.16.2.14*	*Supports 14 hosts Est 3 hosts*
GUESTS	*60*	*172.16.1.0*	*255.255.255.192*	*172.16.1.1 - 172.16.1.62*	*Supports 62 hosts Est 56 hosts*
MANAGEMENT	*99*	*172.16.1.192*	*255.255.255.224*	*172.16.1.193 - 172.16.1.222*	*Supports 30 hosts Est 9 hosts*
VOID	*199*				

1.2 Host Address Assignment Table

Each subnet follows a similar structure, wherein the first three addresses are allocated for the core switch VLAN interface. Note that the third address is used as the shared virtual IP address to apply first hop redundancy. For organization, the HSRP group number is the same as the VLAN number. CORE-SW1 is the active router for the Students, Faculty, Services, and Guests VLAN. Whereas, CORE-SW2 is the active router for the Operations, IT_Office, and Management VLAN. Next, the first ten IP addresses for Students, Instructors, Operations, IT_Office, and the first 3 IP addresses for Guests are excluded from the DHCP Pool to account for scalability. The remaining addresses are allocated to the end devices. DHCP is implemented mostly for PC and mobile devices. Whereas, the IP addresses of servers, printers, and the administrative PC are manually configured. With this, the Services and Management VLAN do not have DHCP pools as their devices are manually configured.

Subnet / VLAN Name: ***STUDENTS***
DHCP Pool Range: *172.16.0.1 - 172.16.0.254* (.1 - .10 excluded)

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 10*	*172.16.0.1*	*255.255.255.0*	*NA*
CORE-SW1	*VLAN 10 (HSRP Group 10)*	*172.16.0.3*	*255.255.255.0*	*NA*
CORE-SW2	*VLAN 10*	*172.16.0.2*	*255.255.255.0*	*NA*
CORE-SW2	*VLAN 10 (HSRP Group 10)*	*172.16.0.3*	*255.255.255.0*	*NA*
1F-TR1-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*
1F-TR2-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*
1F-TR3-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*
2F-TR4-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*
2F-TR5-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*
2F-TR6-SPC1 to PC20	*NIC*	*DHCP-ASSIGNED*	*255.255.255.0*	*172.16.0.3*

Subnet / VLAN Name: ***FACULTY***
DHCP Pool Range: *172.16.1.65 - 172.16.1.126 (.65 - .75 excluded)*

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 20*	*172.16.1.65*	*255.255.255.192*	*NA*
CORE-SW1	*VLAN 20 (HSRP Group 20)*	*172.16.1.67*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 20*	*172.16.1.66*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 20 (HSRP Group 20)*	*172.16.1.67*	*255.255.255.192*	*NA*
1F-TR1-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
1F-TR2-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
1F-TR3-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
2F-TR4-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
2F-TR5-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
2F-TR6-FPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*

2F-FR-FPC1 to FPC10	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.67*
FMobileHost1 to 10	*NIC*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*
WLC	*WLAN-20*	*172.16.1.68*	*255.255.255.192*	*172.16.1.67*

Subnet / VLAN Name: ***OPERATIONS***
DHCP Pool Range: *172.16.1.129 - 172.16.1.190 (.129 - .139 excluded)*

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 30*	*172.16.1.129*	*255.255.255.192*	*NA*
CORE-SW1	*VLAN 30 (HSRP Group 30)*	*172.16.1.131*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 30*	*172.16.1.130*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 30 (HSRP Group 30)*	*172.16.1.131*	*255.255.255.192*	*NA*
1F-AO-OPC1 to OPC11	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.131*
2F-FR-SecPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.131*
2F-LIB-LibrPC1	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.131*
OMobileHost1 to 14	*NIC*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*
WLC	*WLAN-30*	*172.16.1.132*	*255.255.255.192*	*172.16.1.131*

Subnet / VLAN Name: ***IT_OFFICE***
DHCP Pool Range: *172.16.1.225 - 172.16.1.254 (.225 - .235 excluded)*

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 40*	*172.16.1.225*	*255.255.255.224*	*NA*
CORE-SW1	*VLAN 40 (HSRP Group 40)*	*172.16.1.227*	*255.255.255.224*	*NA*
CORE-SW2	*VLAN 40*	*172.16.1.226*	*255.255.255.224*	*NA*
CORE-SW2	*VLAN 40 (HSRP Group 40)*	*172.16.1.227*	*255.255.255.224*	*NA*
1F-ITO-ITPC1 to ITPC5	*NIC*	*DHCP-ASSIGNED*	*255.255.255.224*	*172.16.1.227*
ITMobileHost1 to 5	*NIC*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*	*DHCP-ASSIGNED*
WLC	*WLAN-40*	*172.16.1.228*	*255.255.255.224*	*172.16.1.227*

Subnet / VLAN Name: ***Services***

Device / Host	Port	IP Address	Subnet Mask	Default Gateway

CORE-SW1	*VLAN 50*	*172.16.2.1*	*255.255.255.240*	*NA*
CORE-SW1	*VLAN 50 (HSRP Group 50)*	*172.16.2.3*	*255.255.255.240*	*NA*
CORE-SW2	*VLAN 50*	*172.16.2.2*	*255.255.255.240*	*NA*
CORE-SW2	*VLAN 50 (HSRP Group 50)*	*172.16.2.3*	*255.255.255.240*	*NA*
1F-SR-OrionServer	*NIC*	*172.16.2.6*	*255.255.255.240*	*172.16.2.3*
1F-UR-Printer1	*NIC*	*172.16.2.7*	*255.255.255.240*	*172.16.2.3*
2F-UR-Printer1	*NIC*	*172.16.2.8*	*255.255.255.240*	*172.16.2.3*

Subnet / VLAN Name: ***GUESTS***
DHCP Pool Range: *172.16.1.1 - 172.16.1.62 (.1 to .4excluded)*

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 60*	*172.16.1.1*	*255.255.255.192*	*NA*
CORE-SW1	*VLAN 60 (HSRP Group 60)*	*172.16.1.3*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 60*	*172.16.1.2*	*255.255.255.192*	*NA*
CORE-SW2	*VLAN 60 (HSRP Group 60)*	*172.16.1.3*	*255.255.255.192*	*NA*
2F-LIB-PC1 to PC6	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.3*
GMobileHost1 to 50	*NIC*	*DHCP-ASSIGNED*	*255.255.255.192*	*172.16.1.3*
WLC	*WLAN-60*	*172.16.1.4*	*255.255.255.192*	*172.16.1.3*

Subnet / VLAN Name: ***MANAGEMENT***

Device / Host	Port	IP Address	Subnet Mask	Default Gateway
CORE-SW1	*VLAN 99*	*172.16.1.193*	*255.255.255.224*	*NA*
CORE-SW1	*VLAN 99 (HSRP Group 99)*	*172.16.1.195*	*255.255.255.224*	*NA*
CORE-SW2	*VLAN 99*	*172.16.1.194*	*255.255.255.224*	*NA*
CORE-SW2	*VLAN 99 (HSRP Group 99)*	*172.16.1.195*	*255.255.255.224*	*NA*
1F-SR-AdminPC	*NIC*	*172.16.1.196*	*255.255.255.224*	*172.16.1.195*
1F-SR-AriesServer	*NIC*	*172.16.1.197*	*255.255.255.224*	*172.16.1.195*
1F-TR1-SW1	*VLAN 99*	*172.16.1.198*	*255.255.255.224*	*172.16.1.195*

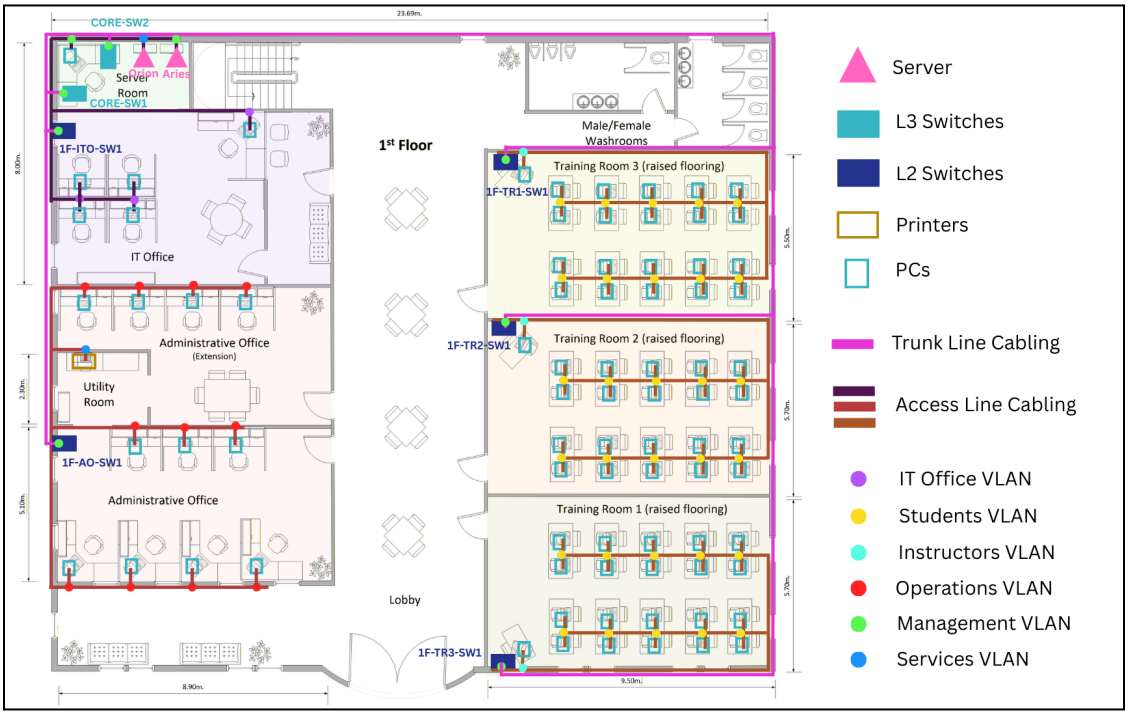
1F-TR2-SW1	*VLAN 99*	*172.16.1.199*	*255.255.255.224*	*172.16.1.195*
1F-TR3-SW1	*VLAN 99*	*172.16.1.200*	*255.255.255.224*	*172.16.1.195*
1F-ITO-SW1	*VLAN 99*	*172.16.1.201*	*255.255.255.224*	*172.16.1.195*
1F-AO-SW1	*VLAN 99*	*172.16.1.202*	*255.255.255.224*	*172.16.1.195*
1F-ITO-AP1	*VLAN 99*	*172.16.1.208*	*255.255.255.224*	*172.16.1.195*
WLC	*VLAN 99*	*172.16.1.209*	*255.255.255.224*	*172.16.1.195*
2F-TR4-SW1	*VLAN 99*	*172.16.1.203*	*255.255.255.224*	*172.16.1.195*
2F-TR5-SW1	*VLAN 99*	*172.16.1.204*	*255.255.255.224*	*172.16.1.195*
2F-TR6-SW1	*VLAN 99*	*172.16.1.205*	*255.255.255.224*	*172.16.1.195*
2F-LIB-SW1	*VLAN 99*	*172.16.1.206*	*255.255.255.224*	*172.16.1.195*
2F-FR-SW1	*VLAN 99*	*172.16.1.207*	*255.255.255.224*	*172.16.1.195*
2F-LIB-AP2	*VLAN 99*	*172.16.1.210*	*255.255.255.224*	*172.16.1.195*

2. Physical Topology

2.1. Floor Plan Illustration

Figure 1

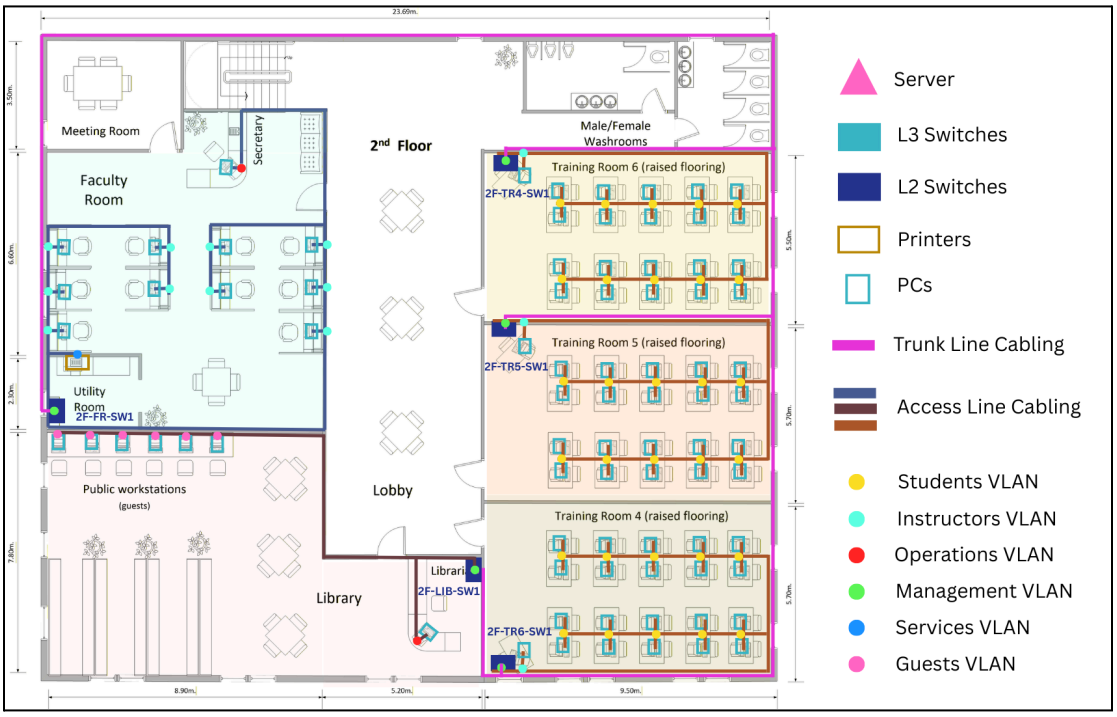
First Floor



The **1F** floor plan in Alonzo IT Training Center (AITC): illustrates the two-tier hierarchical network design. Network equipment was placed across the four departments of the administrative office, IT Office and Training Rooms. The **Server Room** houses two servers (Orion and Aries), two core L3 switches handling all the traffic, one wireless controller, and the admin management PC. This will function as the main distribution frame, where centralized network management takes place.

Figure 2

Second Floor



The other rooms in the 2nd Floor also have their own dedicated access switches: FR-SW1 for the faculty department, LIB-SW1 for the library and guest devices, and TR4-SW1 to TR6-SW1 for the new training rooms. All switches are all placed at an efficient (in latency) and physically secure area of their respective room as wall-mounted switches. All of which are connected to their corresponding L3 switches: CORE-SW1 and CORE-SW2 for manageability. Both floor plans also illustrate that there is structured cabling with patch panels and wall-mounted data cabinet racks. For wireless connection, each floor has one access point and is strategically located in the most central location which is close to the area with the most mobile hosts for a balanced connection in the area. These features provide security, flexibility, and better manageability.

2.2. Device Port Assignment Table

Device: CORE-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
G1/0/1	*1F-TR1-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/2	*1F-TR2-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/3	*1F-TR3-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/4	*1F-ITO-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/5	*1F-AO-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/6	*2F-FR-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/7	*2F-LIB-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/8	*2F-TR4-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
G1/0/9	*2F-TR5-SW1 G0/1*	*Trunk*	*N/A*	*N/A*

G1/0/10	*2F-TR6-SW1 G0/1*	*Trunk*	*N/A*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
G0/1/0/20	*CORE-SW2 G1/0/20*	*Trunk*	*N/A*	*1*
G1/0/21	*CORE-SW2 G1/0/21*	*Trunk*	*N/A*	*1*

Device: 1F-TR1-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*1F-TR1-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*1F-TR1-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*1F-TR1-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*1F-TR1-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*1F-TR1-SPC5*	*Access*	*10*	*N/A*
Fa0/6	*1F-TR1-SPC6*	*Access*	*10*	*N/A*
Fa0/7	*1F-TR1-SPC7*	*Access*	*10*	*N/A*
Fa0/8	*1F-TR1-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*1F-TR1-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*1F-TR1-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*1F-TR1-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*1F-TR1-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*1F-TR1-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*1F-TR1-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*1F-TR1-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*1F-TR1-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*1F-TR1-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*1F-TR1-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*1F-TR1-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*1F-TR1-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*

Fa0/24	*1F-TR1-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW1 G1/0/1*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW2 G1/0/6*	*Trunk*	*N/A*	*N/A*

Device: 1F-TR2-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*1F-TR2-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*1F-TR2-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*1F-TR2-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*1F-TR2-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*1F-TR2-SPC5*	*Access*	*10*	*N/A*
Fa0/6	*1F-TR2-SPC6*	*Access*	*10*	*N/A*
Fa0/7	*1F-TR2-SPC7*	*Access*	*10*	*N/A*
Fa0/8	*1F-TR2-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*1F-TR2-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*1F-TR2-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*1F-TR2-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*1F-TR2-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*1F-TR2-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*1F-TR2-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*1F-TR2-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*1F-TR2-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*1F-TR2-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*1F-TR2-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*1F-TR2-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*1F-TR2-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*1F-TR2-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW1 G1/0/2*	*Trunk*	*N/A*	*N/A*

G0/2	*CORE-SW2 G1/0/7*	*Trunk*	*N/A*	*N/A*

Device: 1F-TR3-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*1F-TR3-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*1F-TR3-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*1F-TR3-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*1F-TR3-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*1F-TR3-SPC5*	*Access*	*10*	*N/A*
Fa0/6	*1F-TR3-SPC6*	*Access*	*10*	*N/A*
Fa0/7	*1F-TR3-SPC7*	*Access*	*10*	*N/A*
Fa0/8	*1F-TR3-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*1F-TR3-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*1F-TR3-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*1F-TR3-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*1F-TR3-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*1F-TR3-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*1F-TR3-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*1F-TR3-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*1F-TR3-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*1F-TR3-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*1F-TR3-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*1F-TR3-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*1F-TR3-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*1F-TR3-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW1 G1/0/3*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW2 G1/0/8*	*Trunk*	*N/A*	*N/A*

Device: 1F-ITO-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannnel group
Fa0/1	*1F-ITO-ITPC1*	*Access*	*40*	*N/A*
Fa0/2	*1F-ITO-ITPC2*	*Access*	*40*	*N/A*
Fa0/3	*1F-ITO-ITPC3*	*Access*	*40*	*N/A*
Fa0/4	*1F-ITO-ITPC4*	*Access*	*40*	*N/A*
Fa0/5	*1F-ITO-ITPC5*	*Access*	*40*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/20	*1F-ITO-AP1 G/0*	*Trunk*	*N/A*	*N/A*
Fa0/21	*WLC G/1*	*Trunk*	*N/A*	*N/A*
Fa0/22	*1F-SR-AriesServer*	*Access*	*99*	*N/A*
Fa0/23	*1F-SR-OrionServer*	*Access*	*50*	*N/A*
Fa0/24	*1F-SR-AdminPC*	*Access*	*99*	*N/A*
G0/1	*CORE-SW1 G1/0/4*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW2 G1/0/9*	*Trunk*	*N/A*	*N/A*

Device: 1F-AO-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel group
Fa0/1	*1F-AO-OPC1*	*Access*	*30*	*N/A*
Fa0/2	*1F-AO-OPC2*	*Access*	*30*	*N/A*
Fa0/3	*1F-AO-OPC3*	*Access*	*30*	*N/A*
Fa0/4	*1F-AO-OPC4*	*Access*	*30*	*N/A*
Fa0/5	*1F-AO-OPC5*	*Access*	*30*	*N/A*
Fa0/6	*1F-AO-OPC6*	*Access*	*30*	*N/A*
Fa0/7	*1F-AO-OPC7*	*Access*	*30*	*N/A*
Fa0/8	*1F-AO-OPC8*	*Access*	*30*	*N/A*
Fa0/9	*1F-AO-OPC9*	*Access*	*30*	*N/A*
Fa0/10	*1F-AO-OPC10*	*Access*	*30*	*N/A*
Fa0/11	*1F-AO-OPC11*	*Access*	*30*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*1F-UR-Printer1*	*Access*	*50*	*N/A*
G0/1	*CORE-SW1 G1/0/5*	*Trunk*	*N/A*	*N/A*

G0/2	*CORE-SW2 G1/0/10*	*Trunk*	*N/A*	*N/A*

Device: CORE-SW2

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel group
G1/0/1	*2F-FR-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/2	*2F-LIB-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/3	*2F-TR4-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/4	*2F-TR5-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/5	*2F-TR6-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/6	*1F-TR1-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/7	*1F-TR2-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/8	*1F-TR3-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/9	*1F-ITO-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
G1/0/10	*1F-AO-SW1 G0/2*	*Trunk*	*N/A*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
G1/0/20	*CORE-SW1 G1/0/20*	*Trunk*	*N/A*	*1*
G1/0/21	*CORE-SW1 G1/0/21*	*Trunk*	*N/A*	*1*

Device: 2F-FR-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel group
Fa0/1	*2F-FR-FPC1*	*Access*	*20*	*N/A*
Fa0/2	*2F-FR-FPC2*	*Access*	*20*	*N/A*
Fa0/3	*2F-FR-FPC3*	*Access*	*20*	*N/A*
Fa0/4	*2F-FR-FPC4*	*Access*	*20*	*N/A*
Fa0/5	*2F-FR-FPC5*	*Access*	*20*	*N/A*
Fa0/6	*2F-FR-FPC6*	*Access*	*20*	*N/A*
Fa0/7	*2F-FR-FPC7*	*Access*	*20*	*N/A*

Fa0/8	*2F-FR-FPC8*	*Access*	*20*	*N/A*
Fa0/9	*2F-FR-FPC9*	*Access*	*20*	*N/A*
Fa0/10	*2F-FR-FPC10*	*Access*	*20*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/23	2F-UR-Printer1	Access	50	N/A
Fa0/24	2F-FR-SecPC1	Access	30	N/A
G0/1	*CORE-SW2 G1/0/1*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW1 G1/0/6*	*Trunk*	*N/A*	*N/A*

Device: 2F-LIB-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel group
Fa0/1	*2F-LIB-PC1*	*Access*	*60*	*N/A*
Fa0/2	*2F-LIB-PC2*	*Access*	*60*	*N/A*
Fa0/3	*2F-LIB-PC3*	*Access*	*60*	*N/A*
Fa0/4	*2F-LIB-PC4*	*Access*	*60*	*N/A*
Fa0/5	*2F-LIB-PC5*	*Access*	*60*	*N/A*
Fa0/6	*2F-LIB-PC6*	*Access*	*60*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/21	*2F-LIB-AP2 G/0*	*Trunk*	*99*	*N/A*
Fa0/24	*2F-LIB-LibrPC1*	*Access*	*30*	*N/A*
G0/1	*CORE-SW2 G1/0/2*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW1 G1/0/7*	*Trunk*	*N/A*	*N/A*

Device: 2F-TR4-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*2F-TR4-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*2F-TR4-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*2F-TR4-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*2F-TR4-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*2F-TR4-SPC5*	*Access*	*10*	*N/A*

Fa0/6	*2F-TR4-SPC6*	*Access*	*10*	*N/A*
Fa0/7	*2F-TR4-SPC7*	*Access*	*10*	*N/A*
Fa0/8	*2F-TR4-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*2F-TR4-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*2F-TR4-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*2F-TR4-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*2F-TR4-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*2F-TR4-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*2F-TR4-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*2F-TR4-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*2F-TR4-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*2F-TR4-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*2F-TR4-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*2F-TR4-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*2F-TR4-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*2F-TR4-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW2 G1/0/3*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW1 G1/0/8*	*Trunk*	*N/A*	*N/A*

Device: 2F-TR5-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*2F-TR5-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*2F-TR5-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*2F-TR5-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*2F-TR5-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*2F-TR5-SPC5*	*Access*	*10*	*N/A*
Fa0/6	*2F-TR5-SPC6*	*Access*	*10*	*N/A*

Fa0/7	*2F-TR5-SPC7*	*Access*	*10*	*N/A*
Fa0/8	*2F-TR5-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*2F-TR5-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*2F-TR5-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*2F-TR5-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*2F-TR5-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*2F-TR5-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*2F-TR5-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*2F-TR5-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*2F-TR5-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*2F-TR5-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*2F-TR5-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*2F-TR5-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*2F-TR5-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*2F-TR5-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW2 G1/0/4*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW1 G1/0/9*	*Trunk*	*N/A*	*N/A*

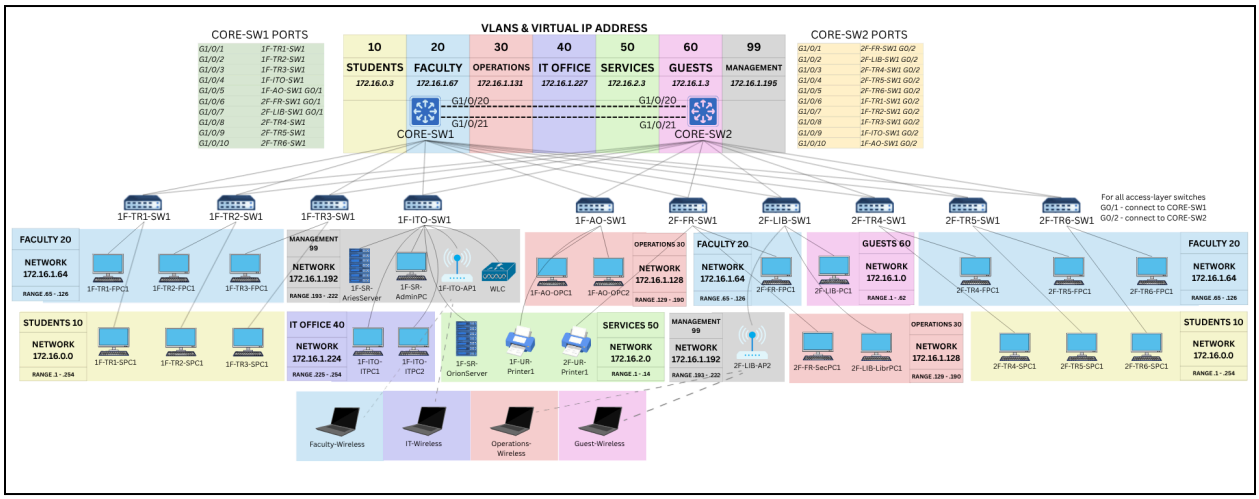
Device: 2F-TR6-SW1

Port	Connected Device	Operational Mode	VLAN (if access)	Etherchannel Group
Fa0/1	*2F-TR6-SPC1*	*Access*	*10*	*N/A*
Fa0/2	*2F-TR6-SPC2*	*Access*	*10*	*N/A*
Fa0/3	*2F-TR6-SPC3*	*Access*	*10*	*N/A*
Fa0/4	*2F-TR6-SPC4*	*Access*	*10*	*N/A*
Fa0/5	*2F-TR6-SPC5*	*Access*	*10*	*N/A*
Fa0/6	*2F-TR6-SPC6*	*Access*	*10*	*N/A*
Fa0/7	*2F-TR6-SPC7*	*Access*	*10*	*N/A*

Fa0/8	*2F-TR6-SPC8*	*Access*	*10*	*N/A*
Fa0/9	*2F-TR6-SPC9*	*Access*	*10*	*N/A*
Fa0/10	*2F-TR6-SPC10*	*Access*	*10*	*N/A*
Fa0/11	*2F-TR6-SPC11*	*Access*	*10*	*N/A*
Fa0/12	*2F-TR6-SPC12*	*Access*	*10*	*N/A*
Fa0/13	*2F-TR6-SPC13*	*Access*	*10*	*N/A*
Fa0/14	*2F-TR6-SPC14*	*Access*	*10*	*N/A*
Fa0/15	*2F-TR6-SPC15*	*Access*	*10*	*N/A*
Fa0/16	*2F-TR6-SPC16*	*Access*	*10*	*N/A*
Fa0/17	*2F-TR6-SPC17*	*Access*	*10*	*N/A*
Fa0/18	*2F-TR6-SPC18*	*Access*	*10*	*N/A*
Fa0/19	*2F-TR6-SPC19*	*Access*	*10*	*N/A*
Fa0/20	*2F-TR6-SPC20*	*Access*	*10*	*N/A*
…	*N/A*	*Shutdown*	*199*	*N/A*
Fa0/24	*2F-TR6-FPC1*	*Access*	*20*	*N/A*
G0/1	*CORE-SW2 G1/0/5*	*Trunk*	*N/A*	*N/A*
G0/2	*CORE-SW1 G1/0/10*	*Trunk*	*N/A*	*N/A*

3. Logical Topology

Figure 3
Logical Topology



4. Security Implementation

4.1. Configured Credentials

Device	Privileged EXEC Password	Console Password	Remote Access Password	VTP Password
CORE-SW1	C0re!gH8%4rT#	Con$C0re!gH8%9	User: netadmin1 Password: N3t!gH8%@kLqZ	^9rT@eX!72$
CORE-SW2	C0re2!gH8%5xZ#	Con$C0re2!gH8%0	User: netadmin2 Password: N3t!gH8%7ZpRm	^9rT@eX!72$
1F-ITO-SW1	IT!sQ2@9zW#4	IT5L!nesQ2@5	User: itadmin Password: IT!sQ2@$cDr	^9rT@eX!72$
1F-AO-SW1	Ops!pX7!8kL#2	Ops$#ine!pX7!9	User: opsadmin Password: Ops!pX7!@wBn	^9rT@eX!72$
1F-TR1-SW1	TR1!tK9#x7Pq$	Tr1$Line!tK9#8	User: tr1admin Password: Tr1!tK9#!QvY	^9rT@eX!72$
1F-TR2-SW1	TR2!a9tK9#Q2z	Tr2$Line!tK9#6	User: tr2admin Password: Tr2!tK9#4mW8	^9rT@eX!72$
1F-TR3-SW1	TR3!tK9#R4mL2	Tr3$Line!tK9#7	User: tr3admin Password: Tr3!tK9#$yF3	^9rT@eX!72$

2F-TR4-SW1	TR1!tK9#x7Pq$	Tr1$Line!tK9#5	User: tr4admin Password: Tr1!tK9#!QvR	^9rT@eX!72$
2F-TR5-SW1	TR2!a9tK9#Q2z	Tr2$Line!tK9#4	User: tr5admin Password: Tr2!tK9#4mW0	^9rT@eX!72$
2F-TR5-SW1	TR3!tK9#R4mL2	Tr3$Line!tK9#3	User: tr6admin Password: Tr3!tK9#$yFTR	^9rT@eX!72$
2F-FR-SW1	FR!a9tK9#Q2z	FR$Line!tK9#2	User: fradmin Password: FR!tK9#4mWT	^9rT@eX!72$
2F-LIB-SW1	LIB!tK9#R4mL2	LIB$Line!tK9#1	User: libadmin Password: LIB!tK9#$yF0	^9rT@eX!72$
WLC	ITadmin	adm!n@Dmln		
	Guest1	g#estPass		
	Faculty1	f#cultyPass		
	Operations1	o#erationsPass		
	ITOffice1	I#officePass		

4.2. Access Control Matrix

		RECEIVER						
		Students (10)	Instructors (20)	Operations (30)	IT (40)	Services (50)	Guests (60)	Management (99)
SENDER	Students		✓	X	✓	X	X	X
	Instructors	✓		✓	✓	✓	X	X
	Operations	X	✓		✓	✓	X	X
	IT	✓	✓	✓		✓	✓	✓
	Services	X	✓	✓	✓		X	X
	Guests	X	X	X	✓	X		X
	Management	X	X	X	✓	X	X	

5. Design Discussion

Alonzo IT Training Center (AITC)'s expansion has required accommodation for a bigger population, a new BYOD policy, and WLAN connectivity. With the use of the retained IPv4 address space (172.16.0.0/22), a VLSM addressing scheme was implemented to meet the growth of users and devices. Five main subnets were expanded from existing departments: Students, Faculty, Operations, IT_Office, Services, Guests, and Management. Moreover, each user group is provided with adequate room for established hosts, while also allowing scalability and future expansion. For example, the 'Students' user

group is assigned a /24 subnet that supports 254 hosts, perfectly providing the established 120 hosts for Students while still allowing room for growth. The rest of the subnets follow accordingly, i.e., Faculty (26 hosts) and Operations (28 hosts) are assigned a /26 subnet, IT_Office (13 hosts) is assigned /27 subnet, and so on.

VLANs were also utilized to segment and isolate traffic between user groups. VLAN 10, 20, 30, 40, 50, 60 for Students, Faculty, Operations, IT_Office, Services, and Guests respectively for their devices and operations. A dedicated VLAN for Management (VLAN 99) is used to house important infrastructure such as SVI management interfaces, servers, and the like. Two additional VLANs are made for unused devices (Void VLAN 199) and untagged traffic (Native VLAN 100). These allow compatibility and easier management throughout the network. Furthermore, two protocols are used to promote availability, efficiency: HRSP and DHCP. As such, CORE-SW1 and CORE-SW2 are configured with HSRP and preemption. A unique physical IP for each VLAN interface and a shared Virtual IP (192.16.0.3) as the default gateway. This is to make sure there is redundancy in the network as it ensures connectivity by eliminating single points of failure. Simply speaking, if the active gateway fails, the standby switch automatically assumes the active role without requiring manual intervention. The core switch will also reclaim the active role once it's operational again.

DHCP is then used in the network to deal with the high volume of user devices by automating the IP address assignment. All department PCs are DHCP-assigned, reducing the risk of overhead and errors in configuration. The first ten addresses in user VLANs and the first three addresses in the Guest VLAN were reserved and excluded from the DHCP pools to allow room for future devices and maintain scalability. Static IP addresses were strictly allocated to servers (Orion and Aries), printers, and the Administrative PC so that these critical devices remain secure and at reachable locations.

The network infrastructure was built using the hierarchical topology established in ITNET02. The topology features two tiers, which is appropriate for a medium-sized office environment with approximately 252 wired and wireless devices. The first tier consists of CORE-SW1 and CORE-SW2, which function as a collapsed core, combining both the core and distribution layers into a single layer. The second tier consists of the remaining switches, which operate as the access layer, providing connectivity to end devices throughout the network.

The infrastructure introduces fault tolerance through the addition of CORE-SW2. Each access layer switch is connected to both core switches using two trunk uplink ports, ensuring that network communication can continue even if one core switch or uplink fails. However, providing an alternate path alone is insufficient to handle the heavy traffic expected from Layer 3 switches. To address this, link aggregation is implemented between CORE-SW1 and CORE-SW2, which improves bandwidth availability, enhances connection reliability, and enables load balancing across the aggregated links.

Additionally, all links connecting intermediary devices have been configured in trunk mode to support the transmission of multiple VLANs across the network. This allows VLAN traffic to traverse the same physical links while maintaining proper traffic segmentation.

At the access layer, switches are allocated to the major rooms of the AITC. This includes one switch for each of the six training rooms, as well as one switch each for the Administrative Office, IT Office, Faculty Room, and Library. These switches provide connectivity to the end-user devices located within their respective areas.

Given the topology and expected volume of traffic, the network is configured with Rapid-PVST+ for a more efficient election and load balancing. The network can easily recalculate paths with less convergence times to the root bridge on any link failures without disruptions and loops. In addition, the redundant links and physical topology present allow for a more deterministic approach when it comes to distributing traffic between core switches. CORE-SW1 is the primary (110) for VLAN 10, 20, 50, and 60 then secondary (100) for VLAN 30, 40, and 99. CORE-SW2 is the opposite: primary for VLAN 30, 40, and 99, while secondary for VLAN 10, 20, 50, and 60. Furthermore, all active access ports on the access layer are configured with PortFast and BPDU Guard.

All devices in the network follow systematic naming conventions based on the format [Floor]-[Room Code]-[Device Identifier], allowing for efficient device identification and better network organization. In terms of physical placement, critical network devices such as the core switches and servers are located in the Server Room on the first floor, while access layer switches are installed in their respective rooms. Service devices, such as printers, are conveniently located in utility rooms on both floors to ensure accessibility for users.

Security and access control is also taken into account by implementing VTP to prevent unauthorized changes to VLANs, remote management using SSH, and having dedicated admin users. All access switches are configured with passwords that fulfills the following strong criteria: more than 8 characters, at least one uppercase character, at least one uppercase character, at least one special character and at least one number.

For wireless connectivity, two access points were added (one on each floor) and they are placed strategically to ensure a strong signal on the floor area. A wireless controller is added as well for centralized management of the wireless network.

6. AI Use Disclosure

6.1. AI Use Reflection

For this project, we used Gemini for the following:

- **Concept clarification:** If some questions were a bit unclear or vague, it was used to give us a better understanding. It was mostly used for suggested implementations for DHCP pools. We made sure to verify this by conducting connectivity tests and using commands like ‘sh ip dhcp pool’ and ‘sh run’. From this, we were able to learn and confirm the usage of a server for DHCP pools and a helper address for the VLAN interfaces of the multilayer-switches. In this implementation, the DHCP pools

will still work regardless if one of the multilayer-switches fail as both of them link to the server using the helper address.

6.2. AI Prompts Used

1. **Concept Clarification**

- "Should dhcp pools be applied to all routers in a network? Is there a way to copy the dhcp pool config to another router automatically"
- "I actually have a Server for Network management. What are the changes I need to make to implement DHCP pools in the network while ensuring redundancy?"